M. Abdullah Ashraf FA23-BCE-120

Maximize $z = -(x_1-\sqrt{5})^2-(x_2-\pi)^2-10$ — (A)

a) Starting point b) Stopping condition

$$X_0 = [6.597 \quad 5.891]^T$$

$$f(x) = -36.597$$

Let stopping condition be $\varepsilon = 0.05$

Iteration 1:

$$x_1 = x_0 + \lambda_1 \nabla f|_{x_0} \quad \text{— (1)}$$

For ∇f

$$\nabla f = \begin{bmatrix} \partial f/\partial x_1 \\ \partial f/\partial x_2 \end{bmatrix} = \begin{bmatrix} -2(x_1-\sqrt{5}) \\ -2(x_2-\pi) \end{bmatrix}$$

$$= \begin{bmatrix} -2(6.597-\sqrt{5}) \\ -2(5.891-\pi) \end{bmatrix}$$

$$\nabla f = \begin{bmatrix} -8.712 \\ -5.499 \end{bmatrix}$$

From eq (1):

$$x_1 = \begin{bmatrix} 6.597 \\ 5.891 \end{bmatrix} + \lambda_1 \begin{bmatrix} -8.722 \\ -5.499 \end{bmatrix}$$

$$\bar{x}_1 = \begin{bmatrix} 6.597-\lambda_1 8.722 \\ 5.891-\lambda_1 5.499 \end{bmatrix} \text{ by combining — (2)}$$

$$f(\bar{x}_1) = -(6.597-8.722\lambda_1-\sqrt{5})^2-(5.891-\lambda_1 5.499-\pi)^2-10$$

$$f(\bar{x}_1) = -(4.301-8.722\lambda_1)-(2.749-5.499\lambda)-10$$

Opening Square

$$f(\bar{x}_1) = -(19.018+76.073\lambda_1^2-76.073\lambda_1)-(7.557+30.8339\lambda_1^2-30.2347\lambda_1)-10$$

$$f(\bar{x}_1) = -106.312\lambda_1^2+106.307\lambda_1-36.575 \quad \text{— (3)}$$

Taking derivative

$f'(\bar{x}_1) = -2(106.312\lambda_1) + 106.307$

Again taking derivative

$f''(\bar{x}_1) = -2(106.312) < 0$

$f(x_1)$ has global maximum at $f'(x_1) = 0$

$f'(\bar{x}_1) = -2(106.312)\lambda_1 + 106.307 = 0$

For λ_1:

$$\lambda_1 = \frac{106.307}{2(106.312)} = 0.5 \quad \text{— (4)}$$

Put in eq (2)

$$\bar{x}_1 = \begin{bmatrix} 6.597 - 8.722(0.5) \\ 5.891 - 5.499(0.5) \end{bmatrix}$$

$$\bar{x}_1 = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix} \quad \text{— (5)}$$

Put in A

$f(\bar{x}_1) - f(\bar{x}_0) = -10.00 + 36.577 = 26.577 > \varepsilon$

Second iteration:

$$\bar{x}_2 = x_1 + \lambda_2 \nabla f|_{x_1} \quad \text{— (6)}$$

$$\bar{x}_1 = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix}$$

$$\nabla f|_{x_1} = \begin{bmatrix} -2(2.236 - \sqrt{5}) \\ -2(3.142 - \pi) \end{bmatrix} = \begin{bmatrix} 0.0001 \\ -0.0008 \end{bmatrix}$$

$$\bar{x}_2 = \begin{bmatrix} 2.236 + 0.0001\lambda_2 \\ 3.142 - 0.0008\lambda_2 \end{bmatrix} \quad \text{— (7)}$$

Scanned with CamScanner

$$f(\bar{x}_2) = -2(2.236+0.0001\lambda_2-\sqrt{5})^2-(3.142-0.0008\lambda_2-\pi)^2-10$$

$$f(\bar{x}_2) = -(-0.000068+0.0001\lambda_2)^2-(0.00041-0.0008\lambda_2)^2-10$$

$$f(\bar{x}_2) = -6.5\times10^{-7}\lambda_2^2+6.696\times10^{-7}\lambda_2-10.000000178 \quad \text{— (8)}$$

Taking derivative

$$f'(\bar{x}_2) = -2(6.5\times10^{-7}\lambda_2)+6.696\times10^{-7}$$

Again taking derivative

$$f''(\bar{x}_2) = -2(6.5\times10^{-7}) < 0$$

$f(x_2)$ has global maximum at $f'(\bar{x}_2)=0$

$$f'(\bar{x}_2) = -2(6.5\times10^{-7}\lambda_2)+(6.696\times10^{-7})=0$$

$$\lambda_2 = \frac{6.696\times10^{-7}}{2(6.5\times10^{-7})} = 0.5151$$

$\boxed{\lambda_2 = 0.5151}$ put in (6)

$$(\bar{x}_2) = \begin{bmatrix} 2.236+(0.0001)(0.5151) \\ 3.142-(0.0008)(0.5151) \end{bmatrix}$$

$$(\bar{x}_2) = \begin{bmatrix} 2.236 \\ 3.142 \end{bmatrix} \quad \text{— (9)}$$

$\bar{x}_2 \cong \bar{x}_1$ $\qquad f(\bar{x}_2) = -10.000$

$|f(\bar{x}_2)-f(x_1)| < \varepsilon$

$x^* = [2.336, 3.142]^T$

$z^* = -10.000$

Original minimization problem

$x^* = [2.236 \;\; 3.142]^T$

$z^* = +10.000.$